

Questions

MathonGo

Q1 - 2024 (04 Apr Shift 2)

If $\int \operatorname{cosec}^5 x dx = \alpha \cot x \operatorname{cosec} x \left(\cos^2 x + \frac{3}{2} \right) + \beta \log_e \left| \tan \frac{x}{2} \right| + C$

where $\alpha, \beta \in \mathbb{R}$ and C is the constant of integration, then the value of $8(\alpha + \beta)$ equals _____

Q2 - 2024 (06 Apr Shift 2)

If $\int \frac{1}{a^2 \sin^2 x + b^2 \cos^2 x} dx = \frac{1}{12} \tan^{-1}(3 \tan x) + \text{constant}$, then the maximum value of $a \sin x + b \cos x$, is :

(1) $\sqrt{40}$

(2) $\sqrt{41}$

(3) $\sqrt{39}$

(4) $\sqrt{42}$

Q3 - 2024 (08 Apr Shift 1)

Let $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$. If $I(0) = 3$, then $I\left(\frac{\pi}{12}\right)$ is equal to

(1) $2\sqrt{3}$

(2) $\sqrt{3}$

(3) $3\sqrt{3}$

(4) $6\sqrt{3}$

Q4 - 2024 (08 Apr Shift 2)

If $\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}} dx = A \left(\frac{\alpha x - 1}{\beta x + 3} \right)^B + C$, where C is the constant of integration, then the value of $\alpha + \beta + 20AB$ is _____

Q5 - 2024 (09 Apr Shift 1)

Let $\int \frac{2 - \tan x}{3 + \tan x} dx = \frac{1}{2}(\alpha x + \log_e |\beta \sin x + \gamma \cos x|) + C$, where C is the constant of integration. Then $\alpha + \frac{\gamma}{\beta}$ is equal to :

(1) 7

Questions

MathonGo

 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
(2) 4						
(3) 1	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
(4) 3						
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo
 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo	 mathongo

Questions

MathonGo

mathongo mathongo mathongo mathongo mathongo mathongo

Answer Key

mathongo mathongo mathongo mathongo mathongo mathongo

Q1 (1) mathongo Q2 (1) mathongo Q3 (3) mathongo Q4 (7) mathongo

Q5 (2) mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

mathongo mathongo mathongo mathongo mathongo mathongo

#MathBoleTohMathonGo

Solutions

MathonGo

Q1

$$\int \operatorname{cosec}^3 x \cdot \operatorname{cosec}^2 x dx = I$$

By applying integration by parts

$$I = -\cot x \operatorname{cosec}^3 x + \int \cot x (-3 \operatorname{cosec}^2 x \cot x \operatorname{cosec} x) dx$$

$$I = -\cot x \operatorname{cosec}^3 x - 3 \int \operatorname{cosec}^3 x (\operatorname{cosec}^2 x - 1) dx$$

$$I = -\cot x \operatorname{cosec}^3 x - 3I + 3 \int \operatorname{cosec}^3 x dx$$

let

$$I_1 = \int \operatorname{cosec}^3 x dx = -\operatorname{cosec} x \cot x - \int \cot^2 x \operatorname{cosec} x dx$$

$$I_1 = -\operatorname{cosec} x \cot x - \int (\operatorname{cosec}^2 x - 1) \operatorname{cosec} x dx$$

$$2I_1 = -\operatorname{cosec} x \cot x + \ln \left| \tan \frac{x}{2} \right|$$

$$I_1 = -\frac{1}{2} \operatorname{cosec} x \cot x + \frac{1}{2} \ln \left| \tan \frac{x}{2} \right|$$

$$4I = -\cot x \operatorname{cosec}^3 x - \frac{3}{2} \operatorname{cosec} x \cot x + \frac{3}{2} \ln \left| \tan \frac{x}{2} \right| + 4c$$

$$I = -\frac{1}{4} \operatorname{cosec} x \cot x \left(\operatorname{cosec}^2 x + \frac{3}{2} \right) + \frac{3}{8} \ln \left| \tan \frac{x}{2} \right| + c$$

$$\therefore \alpha = \frac{-1}{4}, \beta = \frac{3}{8} \rightarrow 8(\alpha + \beta) = 1$$

Q2

$$\int \frac{\sec^2 x dx}{a^2 \tan^2 x + b^2}$$

let $\tan x = t$

$$\sec^2 x dx = dt$$

$$\int \frac{dt}{a^2 t^2 + b^2}$$

$$\frac{1}{a^2} \int \frac{dt}{t^2 + \left(\frac{b}{a}\right)^2}$$

$$\frac{1}{a^2} \frac{1}{\frac{b}{a}} \tan^{-1} \left(\frac{t}{\frac{b}{a}} \right) + c$$

$$\frac{1}{ab} \tan^{-1} \left(\frac{\alpha}{b} \tan x \right) + c$$

$$\text{on comparing } \frac{a}{b} = 3$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

$$ab = 12$$

$$a = 6, b = 2$$

maximum value of

$$6 \sin x + 2 \cos x \text{ is } \sqrt{40}$$

Q3

$$I(x) = \int \frac{6dx}{\sin^2 x (1 - \cot x)^2} = \int \frac{6 \operatorname{cosec}^2 x dx}{(1 - \cot x)^2}$$

$$\text{Put } 1 - \cot x = t$$

$$\operatorname{cosec}^2 x dx = dt$$

$$I = \int \frac{6dt}{t^2} = \frac{-6}{t} + c$$

$$I(x) = \frac{-6}{1 - \cot x} + c, c = 3$$

$$I(x) = 3 - \frac{6}{1 - \cot x}, I\left(\frac{\pi}{12}\right) = 3 - \frac{6}{1 - (2 + \sqrt{3})}$$

$$I\left(\frac{\pi}{12}\right) = 3 + \frac{6}{\sqrt{3} + 1} = 3 + \frac{6(\sqrt{3} - 1)}{2} = 3\sqrt{3}$$

Q4

$$\int \frac{1}{\sqrt[5]{(x-1)^4(x+3)^6}} dx = A \left(\frac{\alpha x - 1}{\beta x + 3} \right)^B + C$$

$$I = \int \frac{1}{(x-1)^{4/5}(x+3)^{6/5}} dx$$

$$I = \int \frac{1}{\left(\frac{x-1}{x+3}\right)^{4/5} (x+3)^2} dx$$

$$\left(\frac{x-1}{x+3}\right) = t \Rightarrow \frac{4}{(x+3)^2} dx = dt \quad t^{-4/5+1}$$

$$I = \frac{1}{4} \int \frac{1}{t^{4/5}} dt = \frac{1}{4} \frac{t^{1/5}}{1/5} + c$$

$$I = \frac{5}{4} \left(\frac{x-1}{x+3}\right)^{1/5} + C$$

$$A = \frac{5}{4} \quad \alpha = \beta = 1 \quad B = \frac{1}{5}$$

$$\alpha + \beta + 20AB = 2 + 20 \times \frac{5}{4} \times \frac{1}{5} = 7$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)

Solutions

MathonGo

Q5

$$\int \frac{2 - \tan x}{3 + \tan x} dx = \int \frac{2 \cos x - \sin x}{3 \cos x + \sin x} dx$$

$$2 \cos x - \sin x = A(3 \cos x + \sin x) + B(\cos x - 3 \sin x)$$

$$3A + B = 2$$

$$A - 3B = -1$$

$$\Rightarrow A = \frac{1}{2}, B = \frac{1}{2}$$

$$\therefore \int \frac{2 \cos x - \sin x}{3 \cos x + \sin x} dx$$

$$= \frac{x}{2} + \frac{1}{2} \ln |3 \cos x + \sin x| + C$$

$$= \frac{1}{2} (x + \ln |3 \cos x + \sin x|) + C$$

$$= \frac{1}{2} (\alpha x + \ln |\beta \sin x + \gamma \cos x|) + C$$

$$\alpha = 1, \beta = 1, \gamma = 3$$

$$\therefore \alpha + \frac{\gamma}{\beta} = 1 + \frac{3}{1} = 4$$

Do you want to practice these PYQs along with PYQs of JEE Main from 2002 till 2024?

[Click here to download MARKS App](#)